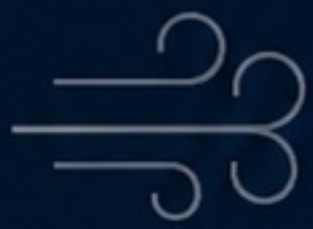


Melissa Exposed the Limit of Wind-Only Categories

StormWise DI² flagged
a compound-risk overload condition.

Category 5 Measures Wind. Real Impact Compounds.

Official hurricane categories are wind-based.
StormWise DI² separately evaluates compound hazard:
terrain amplification, surge residence, and stall retention.



Wind-only category

- Peak sustained wind
- Simple category label
- Does not capture compounding impact

vs



Compound-risk index

- Terrain amplification
- Surge residence
- Stall / retention risk

Compound Flood Risk Index

$$\text{CFRI} = (\text{FTA} \times \text{SMRB}) + \text{SDRL}$$

$$\text{CFRI} = (135.0 \times 0.175) + 156.0$$

$$\text{CFRI} = 179.625$$



Proposed compound Category 6
threshold: **CFRI $\geq$ 178.0**



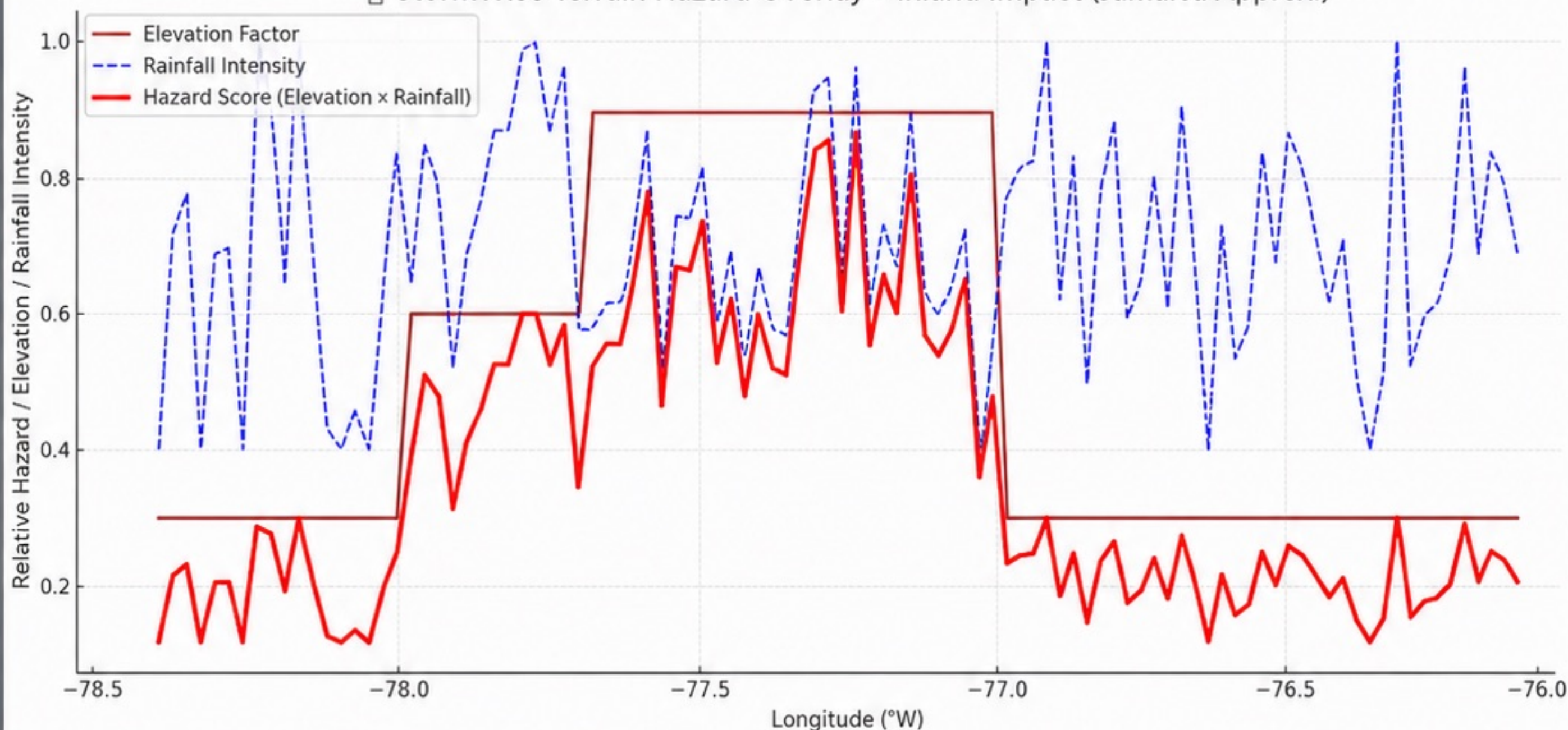
Melissa CFRI:
179.625 – threshold met



Research classification only. Not an official
wind-category designation.

The Risk Was Geographic, Not Just Meteorological

StormWise Terrain Hazard Overlay – Inland Impact (Jamaica Approx.)

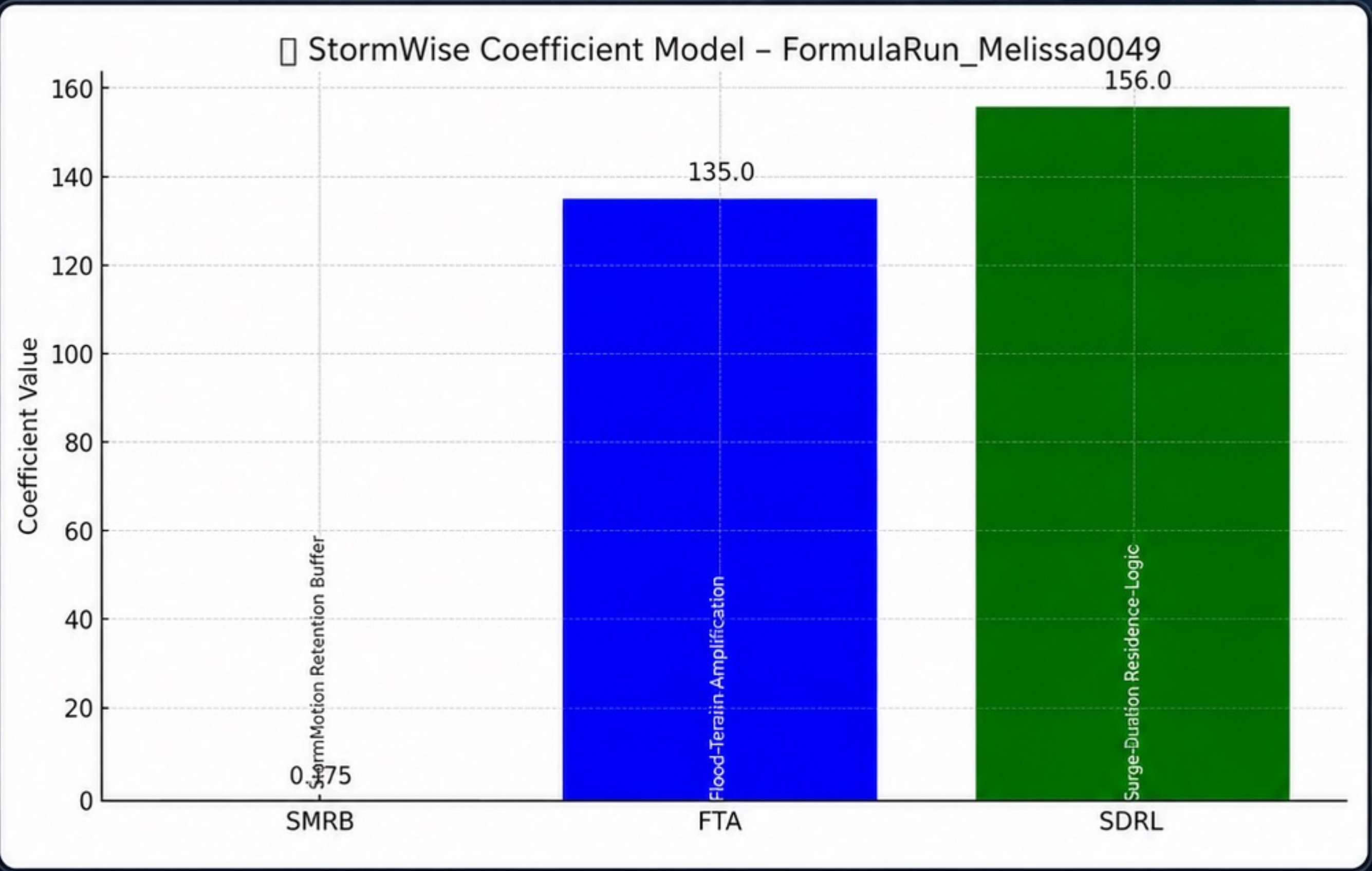


Highest inland amplification appears where rainfall intensity overlaps elevated terrain.

StormWise DI² treated terrain as an **active hazard multiplier**, not background scenery.

FormulaRun_Melissa0049

Coefficient Evidence





FTA =
135.0



SDRL =
156.0



SMRB =
0.175

 Inputs indicate flood-terrain amplification, surge-duration residence, and storm-motion retention.

The Claim Is Narrow — and That's the Point.

- Not official Category 6.
- Not a wind-only classification.
- Not a model-wide performance claim.

StormWise DI² finding:

Melissa met a proposed compound-risk Category 6 threshold under the CFRI framework.

When flood, terrain, surge, and stall risk all redline together, the category should say more than wind.

